

$$1) E_n = \frac{-Z^2 R_H}{n^2} \quad \text{for } \text{He}^+, Z=2 \text{ and } n=1$$

$$E_1 = \frac{-2^2 (13.6 \text{ eV})}{1^2} = -54.4 \text{ eV}$$

$$E = hf \rightarrow 54.4 \text{ eV} (1.602 \times 10^{-19}) = 8.71 \times 10^{-18} \text{ J}$$

$$f = \frac{E}{h} = \frac{8.71 \times 10^{-18} \text{ J}}{6.626 \times 10^{-34} \text{ J}\cdot\text{s}} \approx 1.31 \times 10^{16} \text{ Hz}$$

$$2) E = - \frac{(Z-n)^2 13.6 \text{ eV}}{n^2}$$

$$-5.39 \text{ eV} = - \frac{(3-n)^2 13.6 \text{ eV}}{n^2}$$

$$21.56 = (3-n)^2 13.6$$

$$\sqrt{1.5853} = \sqrt{(3-n)^2}$$

$$1.259 = 3-n$$

$$\boxed{n = 1.741}$$

$$3) \Delta E = \frac{1}{\tau}$$

$$\Delta E = \frac{1.054 \times 10^{-34}}{10^{-8}} = 1.054 \times 10^{-26} \text{ J} = 6.58 \times 10^{-8} \text{ eV}$$

$$E_n = - \frac{13.6 \text{ eV}}{n^2}$$

$$E_1 = -13.6 \text{ eV}$$

$$E_2 = - \frac{13.6}{4} = -3.4 \text{ eV}$$

$$\Delta E_{\text{trans}} = E_1 - E_2 = (-13.6) - (-34) = 10.2 \text{ eV}$$

$$\frac{\Delta E}{\Delta E_1} = \frac{6.58 \times 10^{-8}}{10.2} = 6.45 \times 10^{-9}$$

$$E = \frac{hc}{\lambda} \rightarrow 10.2 (1.602 \times 10^{-19}) = 1.634 \times 10^{-18}$$

$$\lambda = \frac{hc}{E} = \frac{(6.626 \times 10^{-34}) (3 \times 10^8)}{1.634 \times 10^{-18}} = 1.215 \times 10^{-7} \text{ m}$$

$$\boxed{\lambda = 121.5 \text{ nm}}$$

$$\Delta v = \frac{\Delta E}{h} = \frac{1.054 \times 10^{-26}}{6.626 \times 10^{-34}} = 1.59 \times 10^{7} \text{ Hz}$$

$$\lambda = \frac{c}{v}$$

$$\lambda^2 = (1.215 \times 10^{-7})^2 = 1.477 \times 10^{-14} \text{ m}$$

$$\Delta \lambda = \frac{1.477 \times 10^{-14}}{3 \times 10^8} (1.59 \times 10^7) = \boxed{7.83 \times 10^{-16} \text{ m}}$$

$$4) \quad E = \frac{z^2}{2} \quad E_{1s} = \frac{-2^2}{2} = -2$$

$$E_{101} = 2(-2) = -4 \text{ atomic units}$$

$$\psi_{1s}(r) = \sqrt{\frac{2}{\pi}} e^{-2r}$$

$$\psi(r_1, r_2) = \psi_{1s}(r_1) \psi_{1s}(r_2) = \left(\sqrt{\frac{2}{\pi}} e^{-2r_1} \right) \left(\sqrt{\frac{2}{\pi}} e^{-2r_2} \right)$$

$$\boxed{\psi(r_1, r_2) = \frac{2}{\pi} e^{-2(r_1 + r_2)}}$$